

A5: Design Validation Testing (DVT) Report

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The PressMate

A compact, single serving coffee-brewing device

DVT Prototype Overview



Vessel



Plunger



Sleeve

DVT Overview - Vessel

- Body: ***Updated Dimensions***
 - Low-Carbon Steel Round Tube
 - **0.125"** Wall Thickness, **3"** OD
 - Cut to length (**3.5"**) via wire EDM
 - Bore interior to remove weld
 - Turn outside to OD of 2.9"
- Bottom:
 - Steel sheet
 - Thickness **0.188"**
 - Cut to size/shape via water jet
- **Finishing:**
 - Polish with sandpaper and Scotch-Brite
 - Season with grapeseed oil



DVT Overview - Plunger

Plunger

- Body: ***Updated Dimensions***
 - Low-Carbon Steel Round Tube
 - **1/16"** Wall Thickness, **2.75"** OD
 - Cut to length (**4"**) with wire EDM
- Filter:
 - 316 Stainless Steel Wire Cloth
 - 1ft x 1ft Sheet 100 x 100 Mesh Size, 0.006" Opening Size
 - Cut to size with scissors
- O-ring: ***Updated Profile***
 - **X-Profile Silicone**
 - **1/8"** Width, Dash Number **230**
- Polish with sandpaper and Scotch-Brite
- Season with grapeseed oil
- Add silicone lip



DVT Overview - Plunger Lip

- Sorta-Clear 40 Silicone was used to coat the lip of the plunger
- Plungers were dipped into silicone
- A simple tooling fixture was created to hold the plungers to cure



DVT Prototype - Sleeve

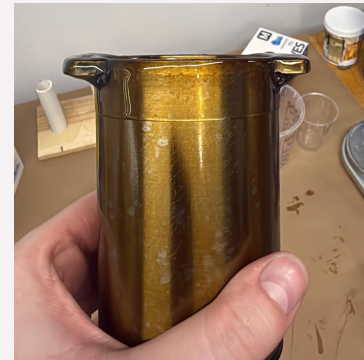
Sleeve

- Two pieces of fabric sandwiching insulation layer
- Sewing machine to put together
- Improved process to only need sewing machine (no hand sewing required)



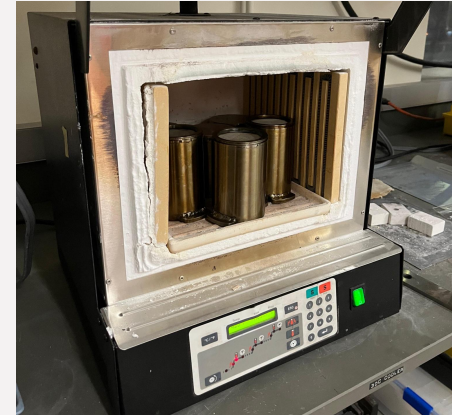
Design Changes

- Increased size of product to meet coffee volume specifications
- X-profile o-ring
 - Provide more surface area while preventing rolling when plunger is pressed through vessel
- Silicone lip
 - Lip of plunger dipped in silicone to provide protection from hot metal



Update to Processes

- Created welding tool to position and secure tabs for plunger
- Seasoned vessel and plunger to create rust-resistant surface
- Added silicone lip to the plunger to protect user from burning lips on hot metal



Risks

- **Achieving consistency:**
 - Out of material quantity capable of producing 6 products, only 3 were usable
 - Vessel boring and notch turning inconsistent due to vessel interior manufacturing weld variation
- **Seasoning:**
 - Difficulty achieving an even coat
- **Silicone Lip:**
 - Difficulty achieving desired silicone consistency and lip thickness

Next Steps

- Order more materials
 - Find food-safe seals
- Adjust silicone process to achieve thicker coat
- Improve process to allow for rapid production